KONGU ENGINEERING COLLEGE

SMART ELEVATOR AUTOMATION AND HUMAN DETECTION SYSTEM USING PIR SENSOR

EXCERSICE - 10

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SMART ELEVATOR AUTOMATION AND HUMAN DETECTION SYSTEM USING PIR SENSOR

1. PROJECT OVERVIEW

The Smart Elevator Automation and Human Detection System is designed to improve elevator safety and automation by detecting the presence of humans using a PIR (Passive Infrared) sensor.

The PIR sensor detects infrared radiation emitted by the human body. When a person approaches the elevator, the sensor sends a signal to the microcontroller. Based on this signal, the controller automatically operates the elevator system.

In the simulation, the elevator operation is represented using output devices such as a servo motor, LEDs, and buzzer. The system demonstrates automatic human detection and elevator operation without requiring continuous manual control.

Main Features

- Automatic human detection
- PIR-based sensing
- Automatic elevator operation
- Visual indication using LEDs
- Audio indication using buzzer
- Servo motor-based elevator/door simulation
- Improved safety and automation

2. PROBLEM STATEMENT

Conventional elevator systems require users to manually operate buttons and may continue operating even when there is no person present.

Manual operation can also result in:

- Unnecessary elevator movement
- Energy consumption

- Delayed operation
- Reduced automation
- Safety issues during entry and exit

Therefore, a simple and cost-effective system is required to detect human presence and automatically control elevator operation.

PROBLEM STATEMENT

"To design and simulate a smart elevator automation and human detection system using a PIR sensor for automatic human presence detection and improved elevator safety and operation."

3. PROPOSED SOLUTION

The proposed system uses a PIR sensor to detect human presence near the elevator.

The PIR sensor continuously monitors the surrounding area. When a human is detected, the sensor sends a HIGH signal to the microcontroller.

The microcontroller then:

1. Detects the person.
2. Activates the elevator operation.
3. Controls the servo motor.
4. Turns ON the appropriate LED.
5. Activates the buzzer for indication.
6. After the required delay, returns the system to the standby condition.

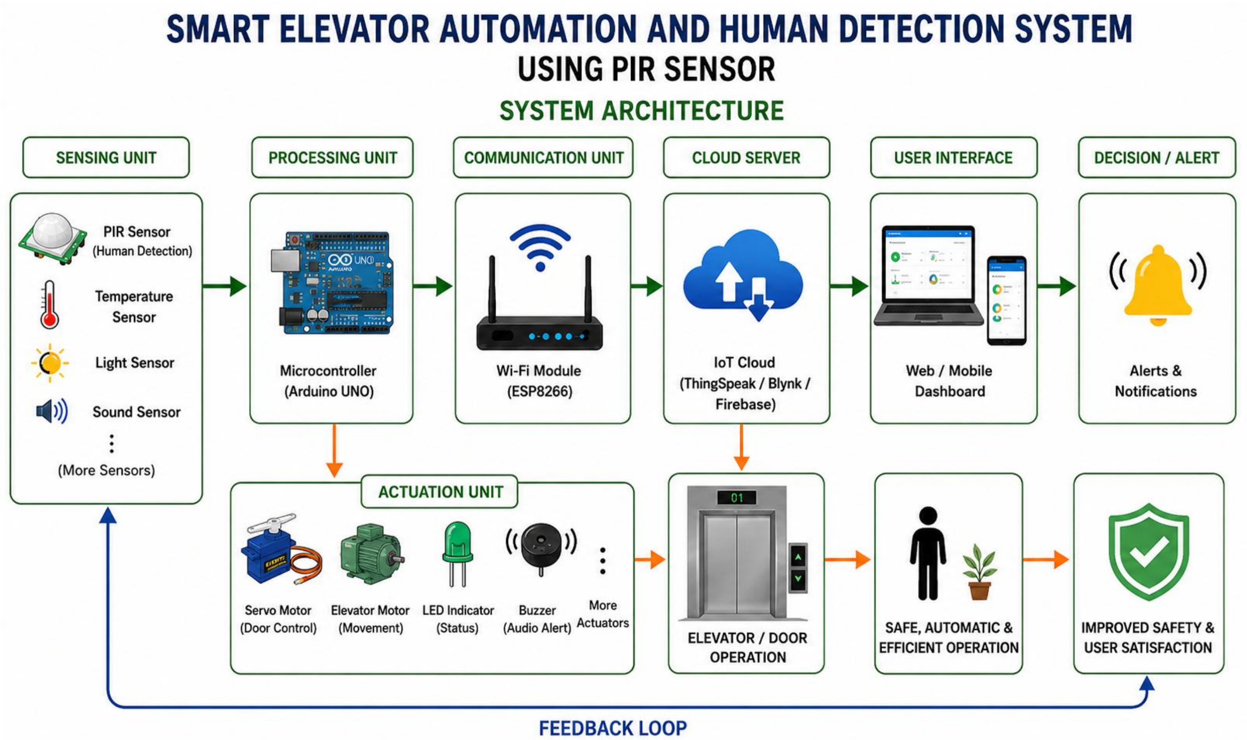
When no human is detected, the elevator remains in the idle condition.

Advantages

- Simple and low-cost
- Automatic operation
- Easy to simulate
- Reduces unnecessary operation
- Improves user convenience
- Can be extended for real elevator applications

4. SYSTEM ARCHITECTURE

The overall architecture of the system is:



5. CIRCUIT TABLE

Component	Pin / Connection	Function
PIR Sensor	VCC	Power supply
PIR Sensor	GND	Ground
PIR Sensor	OUT → Digital Pin	Human detection signal
Servo Motor	VCC	Power supply
Servo Motor	GND	Ground
Servo Motor	Signal → PWM Pin	Motor control
Green LED	Digital Pin	Human detected / operation indication
Red LED	Digital Pin	Standby indication
Buzzer	Digital Pin	Audio indication
Arduino / Controller	USB / Power	Main controller

Example Pin Configuration

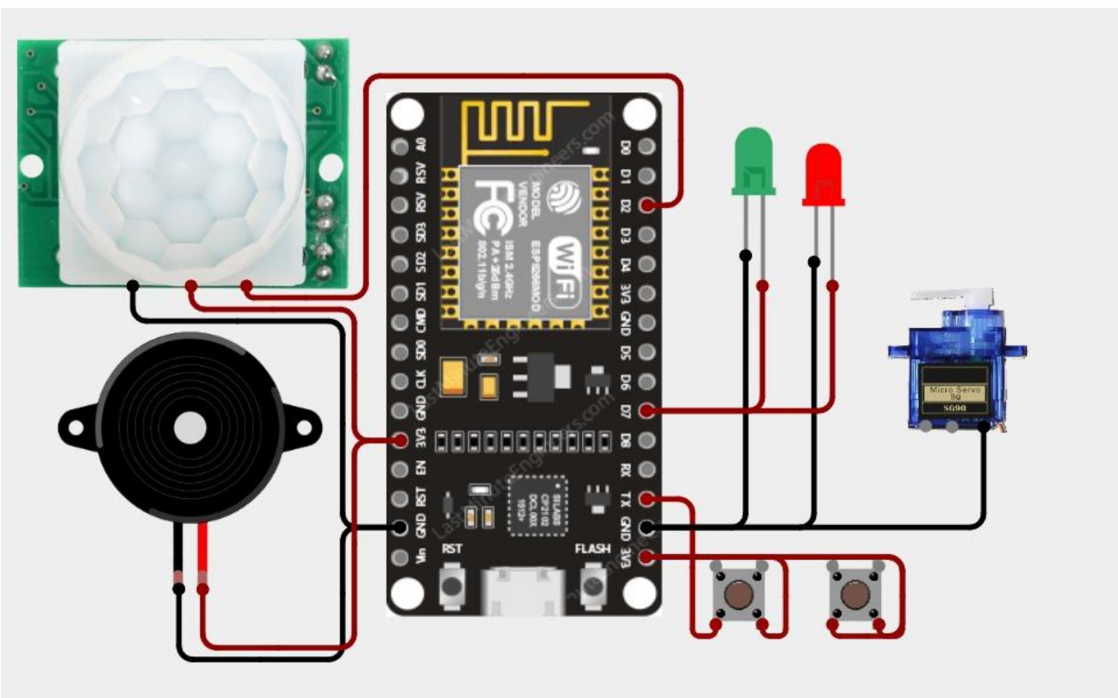
Device	Arduino Pin
PIR OUT	D2
Servo Signal	D9
Green LED	D7
Red LED	D6
Buzzer	D8

6. CIRCUIT DESIGN

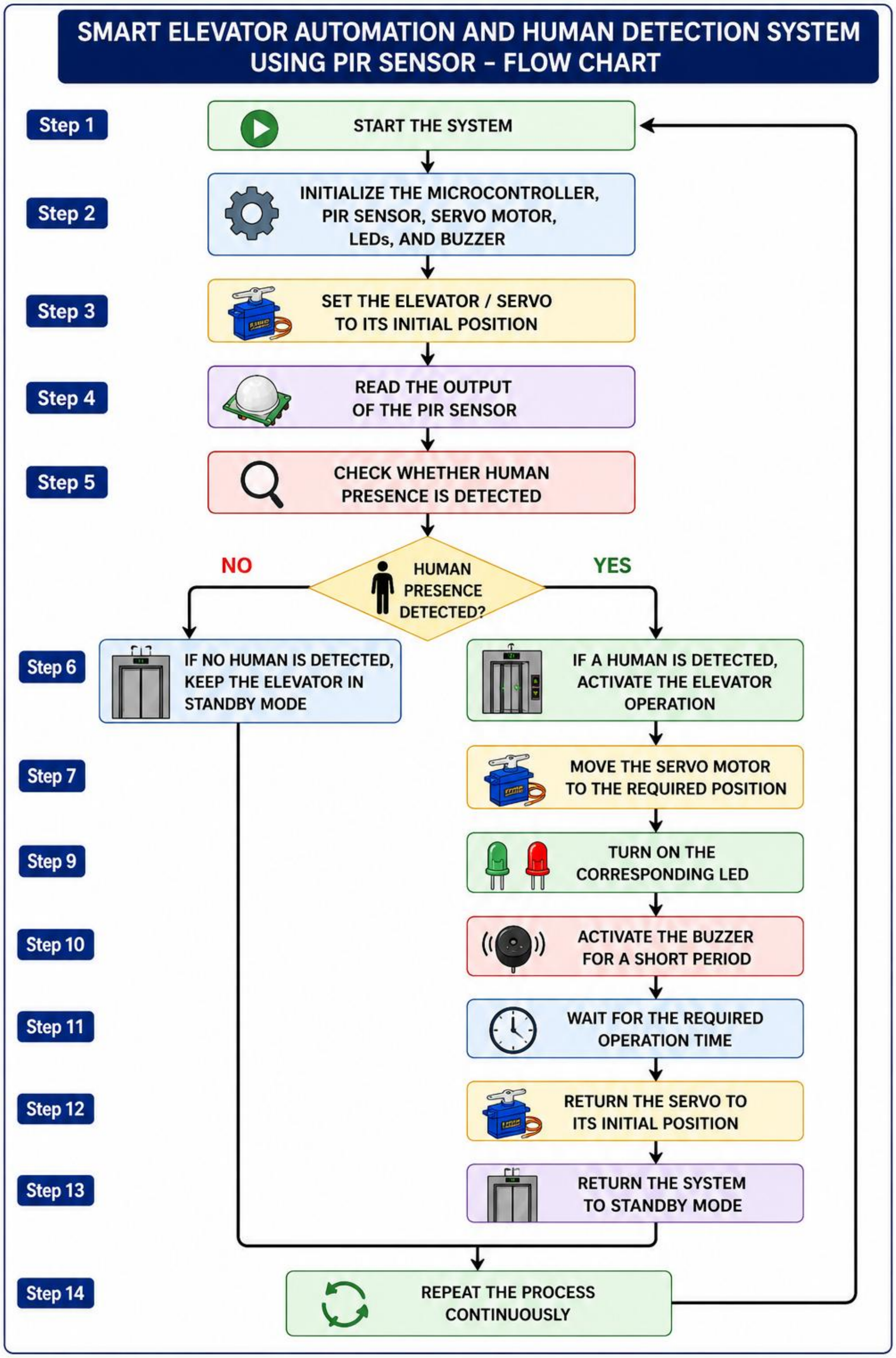
The **PIR sensor** is connected to the power supply and ground. Its output pin is connected to a digital input pin of the microcontroller.

The **servo motor** is connected to a PWM-capable digital pin. The servo represents the elevator door or elevator movement in the simulation.LEDs are connected through suitable resistors to indicate different system conditions.

The buzzer is connected to a digital output pin and provides an audio indication when human presence is detected.



7. ALGORITHM



SMART ELEVATOR AUTOMATION AND HUMAN DETECTION SYSTEM USING PIR SENSOR – FLOW CHART
Step 1
START THE SYSTEM
Step 2
INITIALIZE THE MICROCONTROLLER, PIR SENSOR, SERVO MOTOR, LEDs, AND BUZZER
Step 3
SET THE ELEVATOR / SERVO TO ITS INITIAL POSITION
Step 4
READ THE OUTPUT OF THE PIR SENSOR
Step 5
CHECK WHETHER HUMAN PRESENCE IS DETECTED
HUMAN PRESENCE DETECTED?
NO
YES
Step 6
IF NO HUMAN IS DETECTED, KEEP THE ELEVATOR IN STANDBY MODE
IF A HUMAN IS DETECTED, ACTIVATE THE ELEVATOR OPERATION
Step 7
MOVE THE SERVO MOTOR TO THE REQUIRED POSITION
Step 9
TURN ON THE CORRESPONDING LED
Step 10
ACTIVATE THE BUZZER FOR A SHORT PERIOD
Step 11
WAIT FOR THE REQUIRED OPERATION TIME
Step 12
RETURN THE SERVO TO ITS INITIAL POSITION
Step 13
RETURN THE SYSTEM TO STANDBY MODE
Step 14
REPEAT THE PROCESS CONTINUOUSLY

8. CODING

The following Arduino program demonstrates the basic operation of the smart elevator system.

```
#include <Servo.h>

Servo elevatorServo;

// Pin definitions
const int pirPin = 2;
const int redLED = 6;
const int greenLED = 7;
const int buzzer = 8;
const int servoPin = 9;

void setup()
{
  // Set pin modes
  pinMode(pirPin, INPUT);
  pinMode(redLED, OUTPUT);
  pinMode(greenLED, OUTPUT);
  pinMode(buzzer, OUTPUT);

  // Attach servo
  elevatorServo.attach(servoPin);

  // Initial elevator position
  elevatorServo.write(0);
```

```
  // Initial status
  digitalWrite(redLED, HIGH);
  digitalWrite(greenLED, LOW);
  digitalWrite(buzzer, LOW);

  Serial.begin(9600);

  Serial.println("Smart Elevator System");
  Serial.println("System Ready");
}

void loop()
{
  int pirState = digitalRead(pirPin);

  // Human detected
  if (pirState == HIGH)
  {
    Serial.println("Human Detected!");

    // Change LED indication
    digitalWrite(redLED, LOW);
    digitalWrite(greenLED, HIGH);

    // Buzzer ON
    digitalWrite(buzzer, HIGH);
    delay(500);
```

```
  digitalWrite(buzzer, LOW);

  // Open/activate elevator
  Serial.println("Elevator Activated");

  elevatorServo.write(90);

  delay(3000);

  // Return elevator to initial position
  Serial.println("Elevator Returning");

  elevatorServo.write(0);

  // Return LEDs to standby
  digitalWrite(greenLED, LOW);
  digitalWrite(redLED, HIGH);

  Serial.println("System Ready");

  // Small delay to prevent repeated detection
  delay(2000);
}

// No human detected
else
{
```

```
    digitalWrite(redLED, HIGH);
    digitalWrite(greenLED, LOW);
    digitalWrite(buzzer, LOW);


    elevatorServo.write(0);
  }
}
```

9. SYSTEM WORKING

The system starts by initializing all the connected components.

The PIR sensor continuously monitors the area around the elevator. When a person enters the detection range, the PIR sensor detects the change in infrared radiation and produces a HIGH output signal.

The microcontroller receives this signal and identifies that a person is present.

The controller then:

- Activates the green LED.
- Activates the buzzer.
- Operates the servo motor.
- Represents elevator/door movement.

After a predefined delay, the servo returns to its initial position and the system goes back to standby mode.

The process is continuously repeated for every new human detection.

Operating Conditions

No Human Detected:

- Red LED → ON
- Green LED → OFF
- Buzzer → OFF
- Elevator → Standby

Human Detected:

- Red LED → OFF
- Green LED → ON
- Buzzer → ON temporarily
- Elevator → Activated

10. BILL OF MATERIALS

S.No	Component	Quantity
1	Arduino UNO / Microcontroller	1
2	PIR Sensor	1
3	Servo Motor	1
4	Green LED	1
5	Red LED	1
6	Buzzer	1
7	Resistors	2
8	Breadboard	1
9	Jumper Wires	As required
10	USB / Power Supply	1

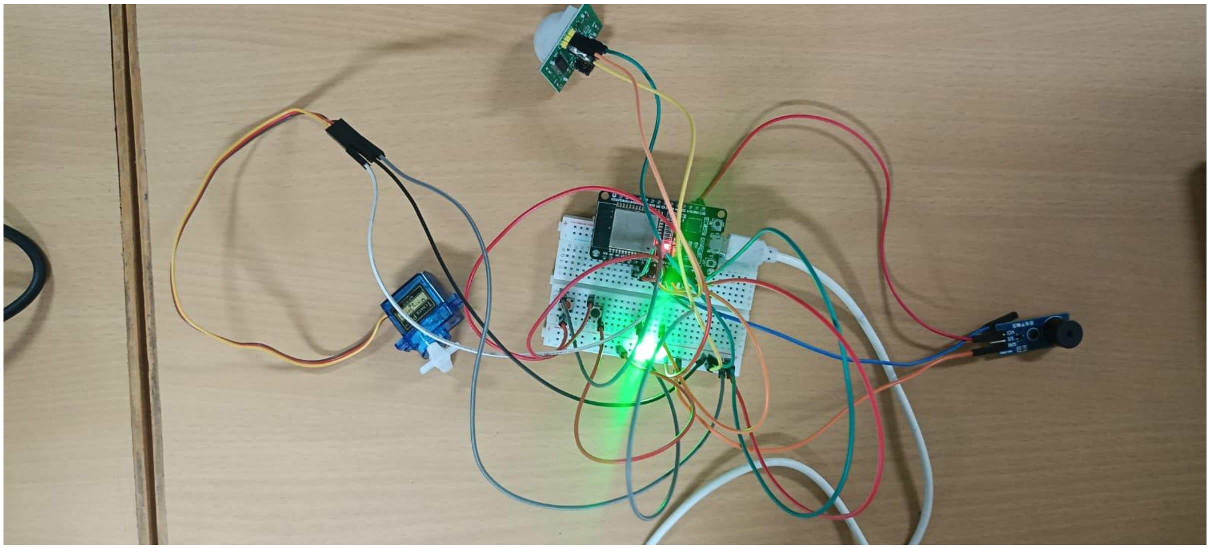
11. PROTOTYPE IMPLEMENTATION

The prototype can be implemented using a microcontroller, PIR sensor, servo motor, LEDs, and buzzer.

The PIR sensor is positioned near the elevator entrance to detect human movement. The microcontroller is connected to all the input and output components.

The servo motor can be mechanically connected to a small elevator door or elevator model. When the PIR sensor detects a person, the servo rotates to represent the opening or activation of the elevator.

The LEDs provide visual information about the system condition, while the buzzer provides an audio indication.

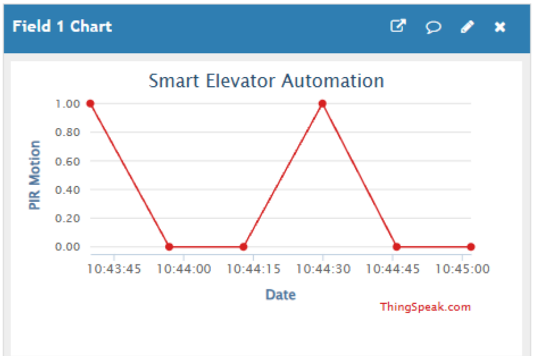


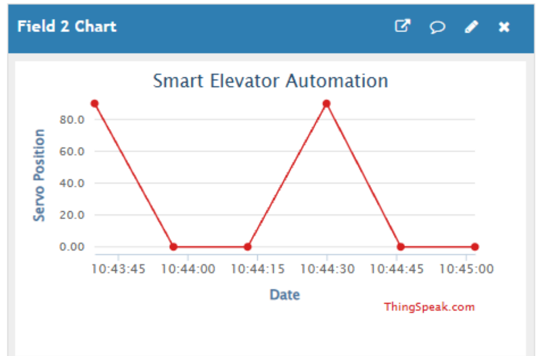
12. RESULTS & PERFORMANCE ANALYSIS

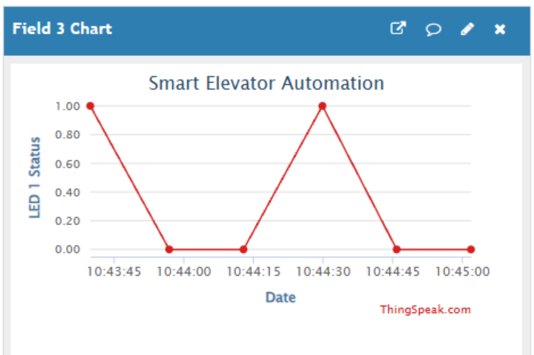
The proposed system successfully demonstrates automatic human detection and elevator operation using a PIR sensor.

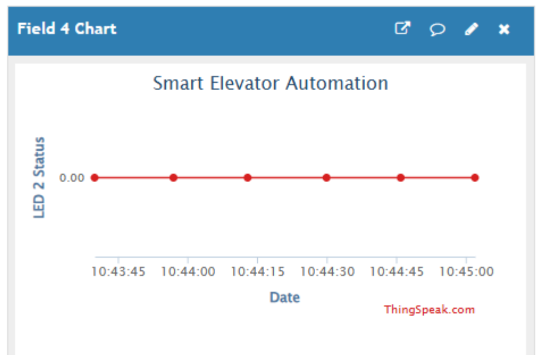
Expected Results:

Test Condition	PIR Output	Elevator	LED	Buzzer
No human	LOW	Standby	Red ON	OFF
Human detected	HIGH	Activated	Green ON	ON
After operation	LOW	Returns to standby	Red ON	OFF









Performance Analysis

The system provides:

- Reliable human presence detection within the PIR sensor's detection range.
- Automatic elevator operation.
- Quick response to human movement.
- Simple control using a microcontroller.
- Low-cost implementation.
- Easy integration with additional sensors and safety systems.

Conclusion

The Smart Elevator Automation and Human Detection System using PIR Sensor successfully demonstrates the basic concept of an automated elevator system.

The PIR sensor detects human presence and sends the information to the microcontroller. The controller automatically operates the servo motor and provides LED and buzzer indications.

This system can be further improved by adding multiple floors, floor selection buttons, ultrasonic sensors, door obstruction detection, LCD display, emergency stop, load sensors, and IoT monitoring.

Thus, the proposed system provides a simple foundation for developing a safer, smarter, and more automated elevator system.